



PowerPoint Presentation by
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MANAGEMENT ACCOUNTING

8th EDITION

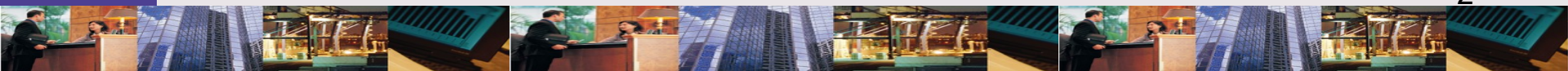
BY

HANSEN & MOWEN

**16 LEAN ACCOUNTING, TARGET
COSTING, & BALANCED SCORECARD**

LEARNING OBJECTIVES

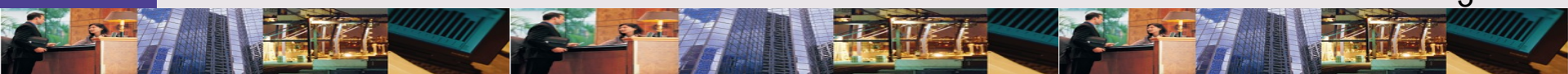
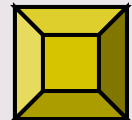
After studying this chapter, you should be able to:



LEARNING OBJECTIVES

1. Describe the basic features of lean manufacturing.
2. Describe lean accounting.
3. Explain the basics of life-cycle cost management & target costing.
4. Discuss the basic features of the Balanced Scorecard & its role in lean manufacturing.

*Click the button to skip
Questions to Think About*



QUESTIONS TO THINK ABOUT: Allen Autoparts, Inc.

How does lean manufacturing
change cost accounting &
management?

QUESTIONS TO THINK ABOUT: Allen Autoparts, Inc.

What are the similarities
between JIT & lean
manufacturing?

QUESTIONS TO THINK ABOUT: Allen Autoparts, Inc.

How are products assigned costs in a lean manufacturing environment?

QUESTIONS TO THINK ABOUT: Allen Autoparts, Inc.

Why are processes so important
to performance management?

QUESTIONS TO THINK ABOUT: Allen Autoparts, Inc.

Are lean manufacturing and the
Balanced Scorecard compatible
approaches?

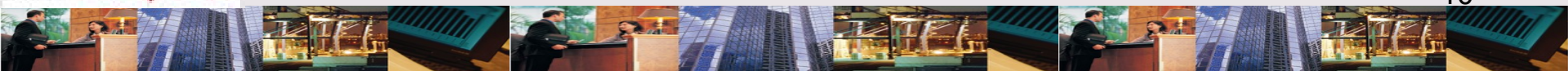
LEARNING OBJECTIVE

1

Describe the basic features of lean manufacturing.

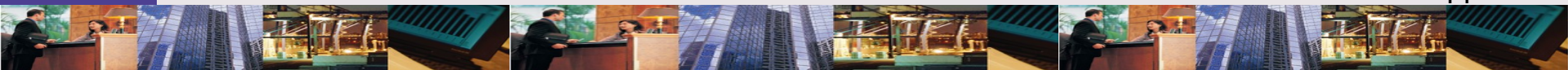
ALLEN AUTOPARTS: Background

Allen Autoparts is concerned about competition in an environment that changes rapidly. They need to exercise better control, reduce costs, become more efficient, and gain operating efficiencies. Can lean manufacturing help?



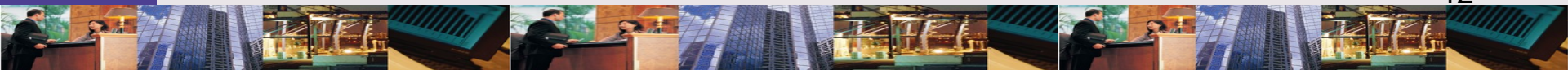
LEAN MANUFACTURING: Definition

Is an approach designed to eliminate waste & maximize customer value.



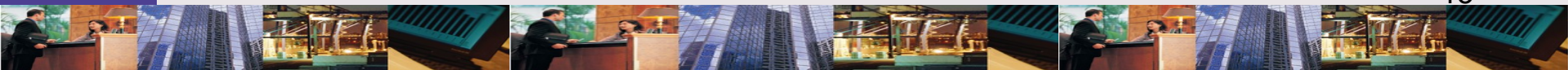
DIMENSIONS OF LEAN MANUFACTURING

- ✧ Delivering the right product
 - ✧ Right quantity
 - ✧ Right quality (zero defect)
 - ✧ At time needed
 - ✧ At lowest possible cost
- ✧ A cost reduction strategy that redefines activities performed



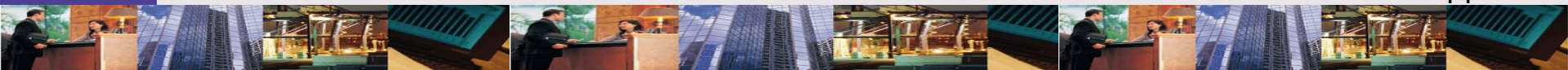
5 PRINCIPLES OF LEAN THINKING

1. Precisely specify value by each particular product
2. Identify the “value stream” for each
3. Make value flow without interruption
4. Let customer pull value from producer
5. Pursue perfection



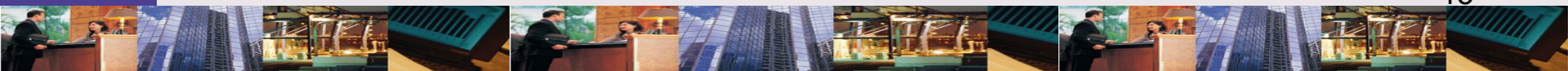
VALUE BY PRODUCT: Definition

Is when only **value-added features** should be produced; **non-value-added activities** should be eliminated.



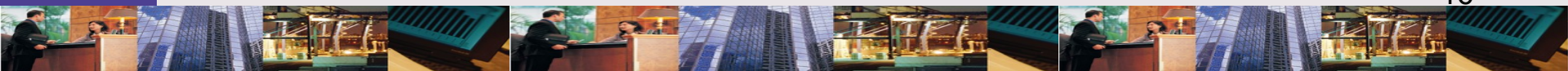
VALUE STREAM: Definition

Is all activities, **both value-added & non-value-added**, required to bring product group or service from starting point to finished product in hands of customer.



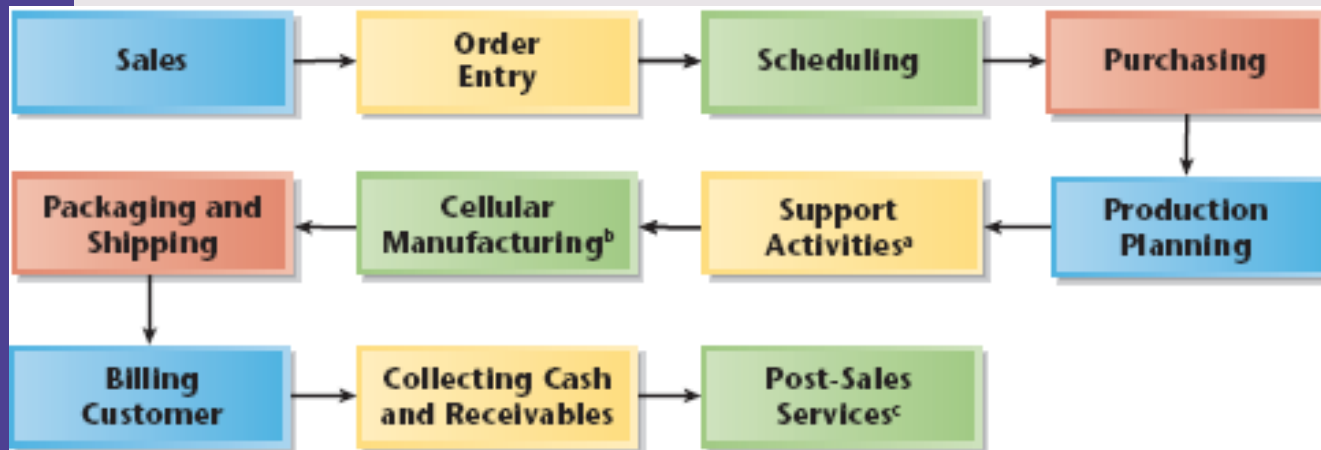
VALUE STREAM

- ✧ Types of value streams
 - ✧ Order fulfillment
 - ✧ New product
- ✧ Value stream activities
 - ✧ Non-value-added
 - ✧ Activities avoidable in the short run
 - ✧ Unavoidable activities due to current technology or production method
 - ✧ Value added



ORDER FULFILLMENT VALUE STREAM

Order fulfillment provides current products to current customers.



^aMoving materials, quality management, engineering, setting up equipment, maintenance, etc.

^bCutting, drilling and insertion, assembly, and finishing.

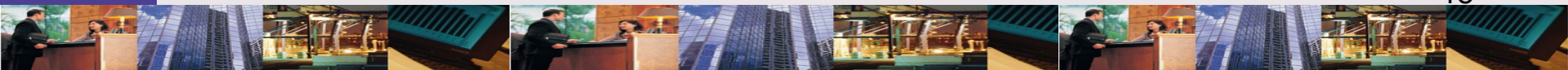
^cCustomer complaints, field repairs, warranty services, etc.

EXHIBIT 16-1

VALUE FLOW

Changes the traditional manufacturing setup for batches to a cellular approach in order to:

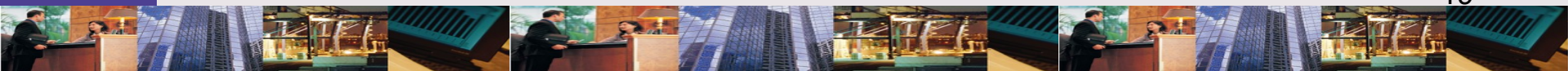
- ✧ Reduce setup time
- ✧ Reduce changeover time



MANUFACTURING CELL:

Definition

Contains all operations in close proximity that are needed to produce a family of products.



TRADITIONAL BATCH SYSTEM

Note time lost in moving & waiting.

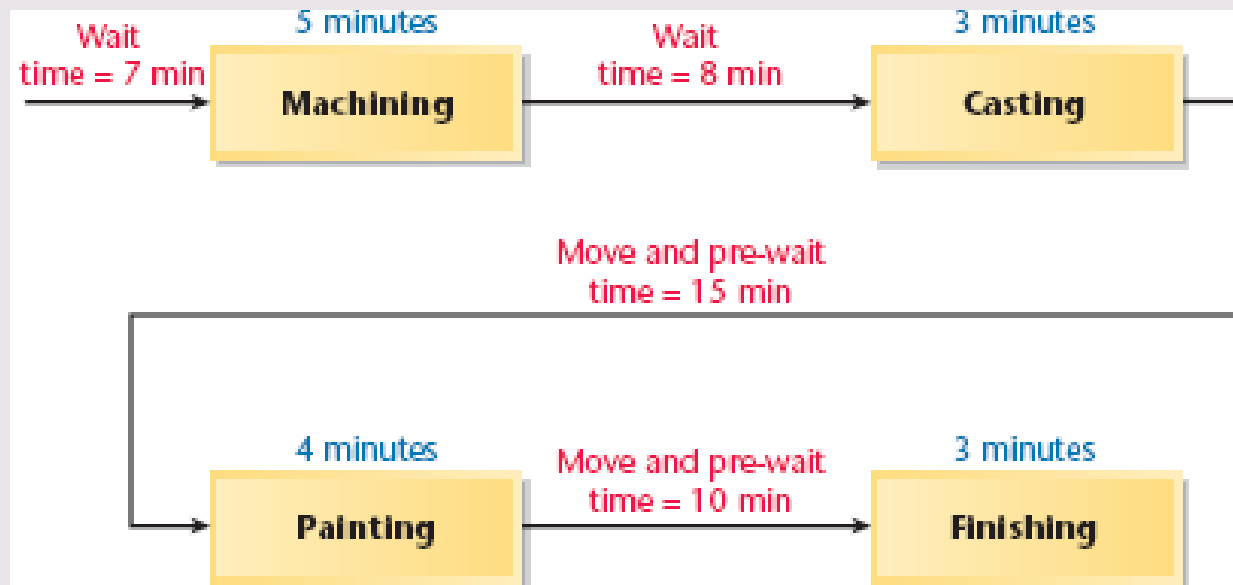
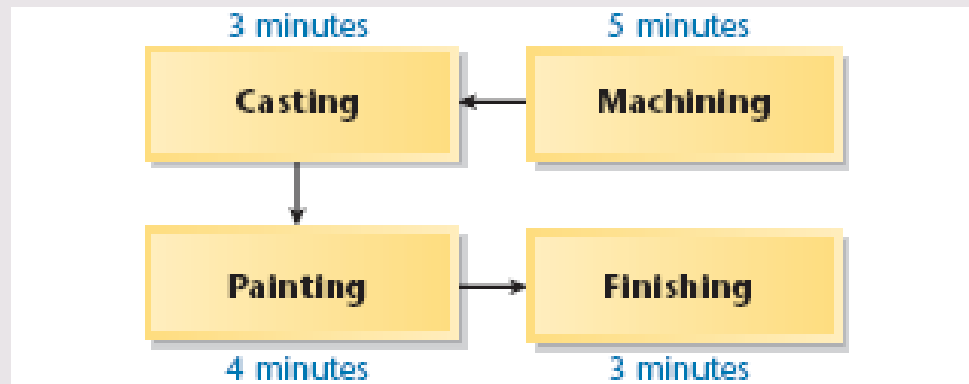


EXHIBIT 16-3A

CELLULAR SYSTEM



Time saved
over traditional
manufacturing
is 90 minutes
(150 – 60).

EXHIBIT 16-3B

PULL VALUE

Lean manufacturing uses a *demand pull* system to reduce waste.

- ✧ JIT inventory

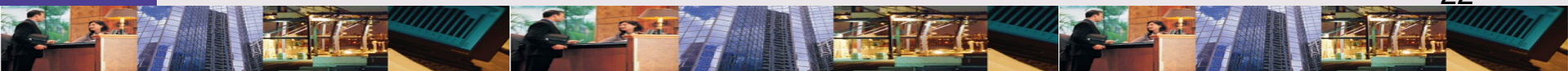
 - ✧ Reduces inventory levels

 - ✧ Requires close relations with suppliers

- ✧ Suppliers benefit from

 - ✧ Long term relations

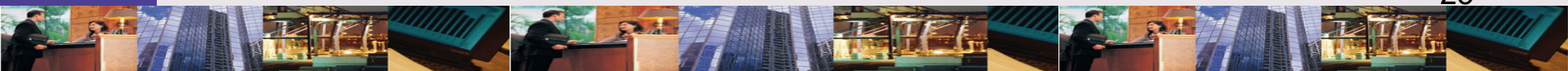
 - ✧ Better competitive position



LEARNING OBJECTIVE

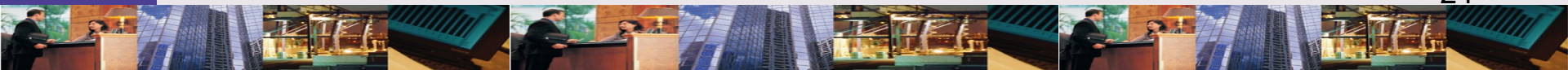
2

Describe lean accounting.



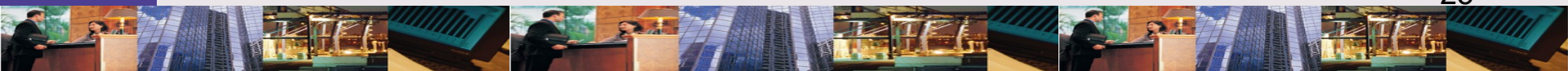
LEAN ACCOUNTING: A Comparison

Traditional cost management systems may not be compatible with Lean Accounting. Lean Accounting makes product costs more simple & direct. More labor and overhead costs are assigned to products through **direct tracing** rather than allocation.



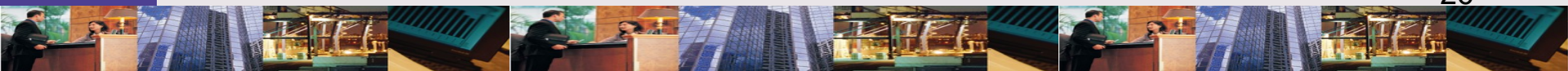
FOCUSED VALUE STREAMS: Definition

Allow overhead costs to be assigned through **driver tracing** of costs in a lean accounting system.



FOCUSED VALUE STREAMS

- ✧ Are more simple & accurate in product costing
- ✧ Have limitations
 - ✧ Initially, labor costs may be difficult to assign if people are employed in several value streams
 - ✧ Labor costs should assigned proportionately
- ✧ Are organized around a family of products



FORMULA: Multiple Products

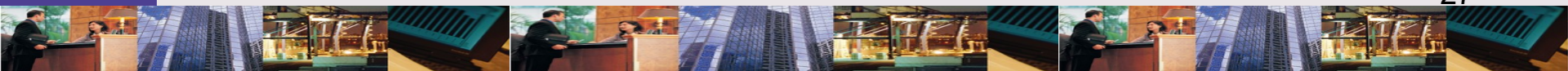
Costs are assigned proportionately when multiple products are produced.

Value stream product cost:

= Total value stream cost of period

÷ Units shipped of period

= \$600,000 / 5,000 = \$120 per unit



VALUE STREAM REPORTING

Costs are collected, reported by value stream; outside costs reported separately.

Allen Autoparts			
	Aluminum Stream	Steel Stream	Plant Totals
Revenues	\$ 700,000	\$1,500,000	\$2,200,000
Material costs	(280,000)	(410,000)	(690,000)
Conversion costs	<u>(70,000)</u>	<u>(190,000)</u>	<u>(260,000)</u>
Value stream profit	\$ 350,000	\$ 900,000	\$1,250,000
Value stream ROS ^a	50%	60%	
Employee costs			(40,000)
Other expenses			(30,000)
Change in inventory:			
Current less prior period			<u>(500,000)</u>
Plant gross profit			<u>\$ 680,000</u>
Plant ROS			31%

^aROS = Return on Sales = Profit/Sales

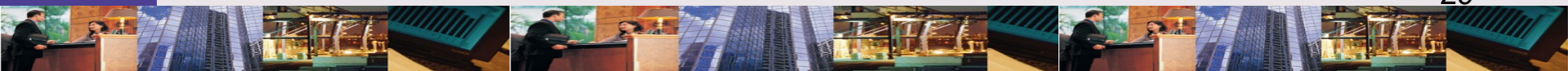
Exhibit 16-6 Allen Autoparts—Weekly Profit & Loss Statement



EXHIBIT 16-6

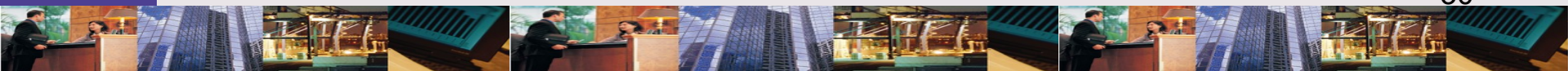
VALUE STREAM DECISIONS

- ✧ May lead to
 - ✧ Short term decisions
 - ✧ May not reflect long term consequences



PERFORMANCE MEASUREMENT: A Comparison

Lean accounting replaces standard cost system measurements with a Box Scorecard that compares a) operational, b) capacity, & c) financial metrics with prior week performances. A mixture of financial & nonfinancial measures are used.



BOX SCORECARD

	Last Week	This Week (4/6/08)	Planned Future State (6/30/08)
Operational			
Units per person	250	270	280
On-time delivery	90%	92%	97%
Dock-to-dock days	18.5	18	16
First time through	56%	58%	65%
Average product cost	\$128	\$120	\$115
Accounts receivable days	31	30	28
Capacity			
Productive	21%	20%	25%
Nonproductive	45%	46%	30%
Available	34%	34%	45%
Financial			
Weekly sales	\$1,800,000	\$1,500,000	\$2,000,000
Weekly material cost	\$800,000	\$600,000	\$600,000
Weekly conversion	\$400,000	\$300,000	\$400,000
Weekly value stream profit	\$600,000	\$600,000	\$1,000,000
ROS	33%	40%	50%

Comparison measures point to future desired goals.

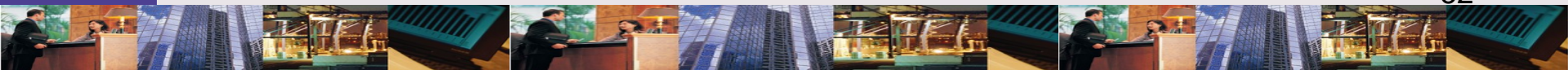
Exhibit 16-7 Value-Stream Box Scorecard for 4/6/2008

EXHIBIT 16-7

LEARNING OBJECTIVE

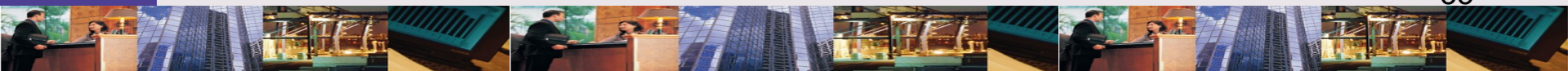
3

Explain the basics
of life-cycle cost
management &
target costing.



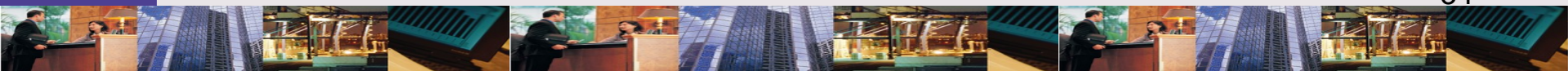
What are product life cycle & life cycle costs?

Product life cycle is the time a product exists from conception to abandonment. **Life cycle costs** are all costs associated with a product for its life cycle.



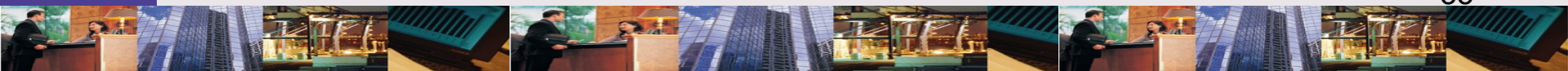
VALUE CHAIN: Definition

Is the set of activities required to design, develop, produce, market, and service a product.



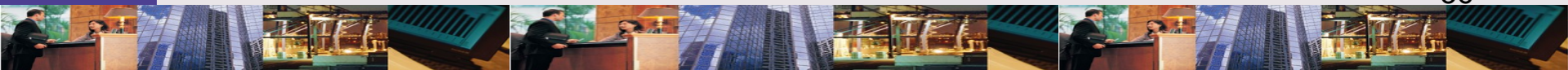
When are most costs incurred?

During the **development stage**.
This is also the time costs should best be managed.



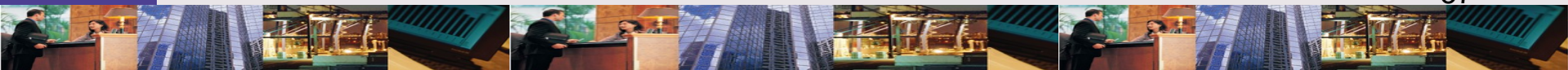
WHOLE-LIFE PRODUCT COST

- ✧ Product cost is
 - ✧ Nonrecurring costs
 - ✧ Planning,
 - ✧ Designing,
 - ✧ Testing
 - ✧ Manufacturing costs
 - ✧ Logistic costs
 - ✧ Customer's postpurchase costs



TARGET COST: Definition

Is the difference between sales price needed to capture a predetermined market share & desired per-unit profit.



TARGET COSTING

Uses 1 of 3 methods

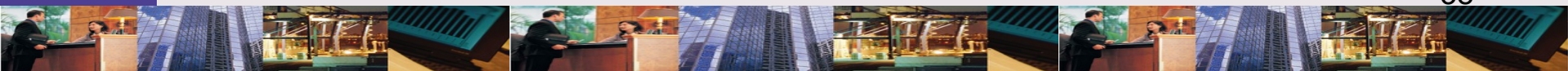
- ✧ Reverse engineering

 - ✧ Tearing down a competitors product to discover design features that create cost reductions

- ✧ Value analysis

 - ✧ Attempting to assess the value placed on product functions by customers

- ✧ Process improvement



TARGET COSTING MODEL

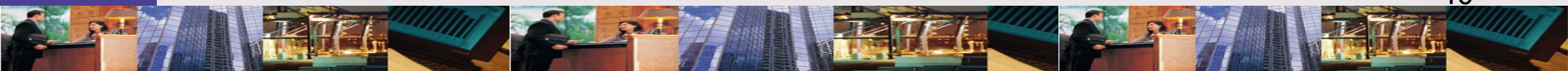


When desired profit not met, target product costing to redesign product, process.

OTHER ISSUES

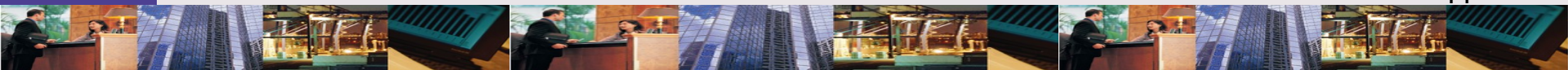
✧ Short life cycles

✧ Life cycle cost management even more important when life cycle is short



LIFE CYCLE COSTING: A Comparison

Life cycle costing includes development costs unlike conventional cost systems. Inclusion of more cost information can be useful for assessing effects on costs and benefit future design.



PERFORMANCE REPORT: Life Cycle Costing

Variances are computed between actual & budgeted costs.

Year	Item	Actual Costs	Budgeted Costs	Variance
2008	Development	\$190,000	\$200,000	\$10,000 F
2009	Production	300,000	240,000	60,000 U
	Logistics	75,000	80,000	5,000 F
2010	Production	435,000	360,000	75,000 U
	Logistics	110,000	120,000	10,000 F

*Analysis: Production costs were higher than expected because insertions of diodes and integrated circuits also drive costs (both production and postpurchase costs). Conclusion: The design of future products should try to minimize total insertions. (Postpurchase costs are costs incurred by the customer and so would not be included in the budgeted income statements.)

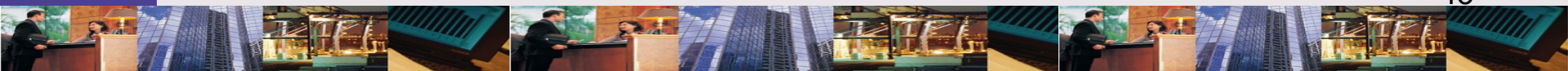
Exhibit 16-11 Performance Report: Life-Cycle Costs*

EXHIBIT 16-11

LEARNING OBJECTIVE

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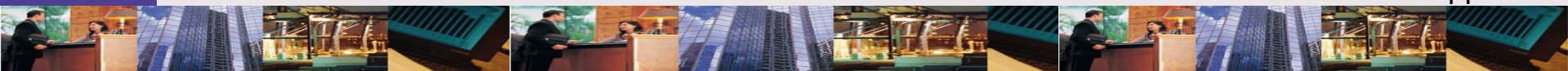
Discuss the basic features of the Balanced Scorecard & its role in lean manufacturing.



BALANCED SCORECARD:

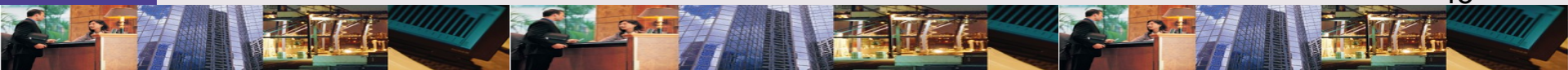
Definition

Translates an organization's mission & strategy into operational objectives & performance measures.



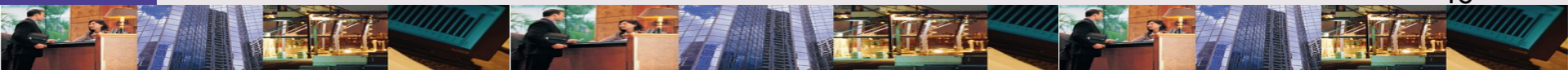
BALANCED SCORECARD PERSPECTIVES

- ✧ Financial perspective
 - ✧ Economic consequences of actions taken in other 3 perspectives
- ✧ Customer perspective
 - ✧ Defines customer & market segments where the business unit will compete
- ✧ Internal business process perspective
 - ✧ Describes internal processes needed to provide value for customers, owners
- ✧ Learning & growth (infrastructure) perspective
 - ✧ Defines capabilities that an organization must have to create long term growth & improvement

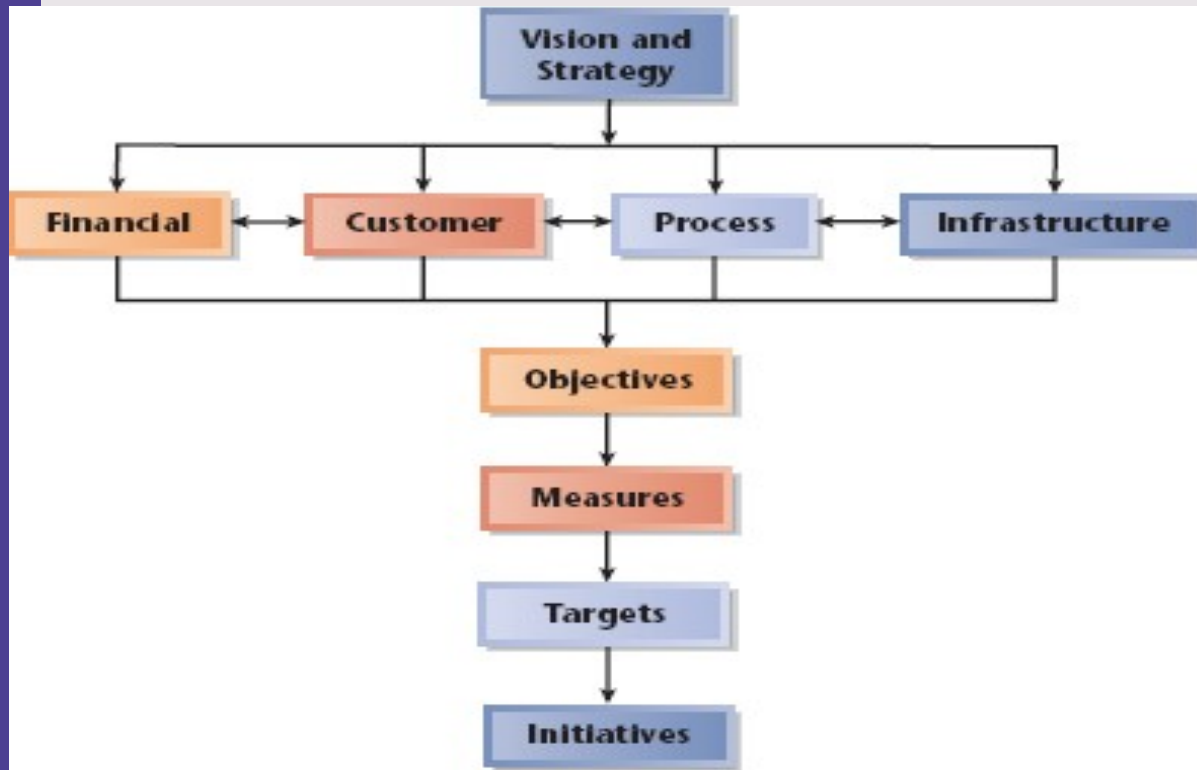


STRATEGY + TRANSLATION

Is the ways in which a company implements its strategy for profit & growth within the balanced scorecard framework. It includes choices of type of customer, product, market, internal & business processes, etc. Strategy translation means specifying objectives, measures, targets & initiatives.



STRATEGY TRANSLATION PROCESS



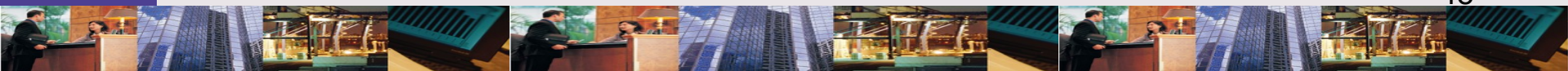
Vision & strategy works through 4 perspectives to reach targets & initiatives.

EXHIBIT 16-12

PERFORMANCE MEASURES

Must be balanced between:

- ✧ Lead measures (performance drivers)
- ✧ Lag (outcome) measures
- ✧ Objective (quantifiable & verifiable) measures
- ✧ Subjective (more judgmental) measures
- ✧ Financial & nonfinancial measures
- ✧ External & internal measures



LINKING PERFORMANCE MEASURES & STRATEGY

✧ Testable strategy

- ✧ Using cause & effect

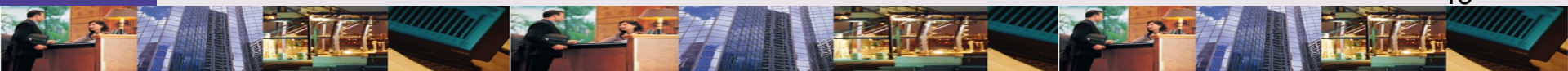
- ✧ Link objectives to overall goal

- ✧ Double loop feedback

 - ✧ Managers receive information on effectiveness of strategy & its underlying assumptions

- ✧ Single loop feedback

 - ✧ Emphasizes only effectiveness of strategy



TESTABLE STRATEGY

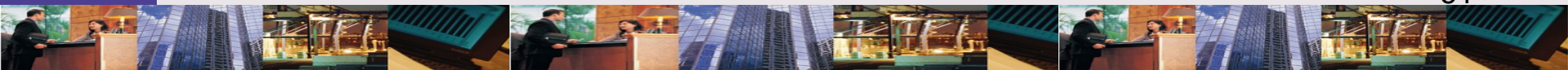
Strategy map illustrates quality improvement strategy.



EXHIBIT 16-13

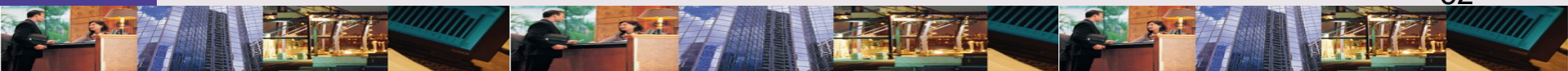
FINANCIAL PERSPECTIVE

- ✧ Flows from other 4 perspectives
 - ✧ Revenue growth
 - ✧ Cost reduction
 - ✧ Asset utilization



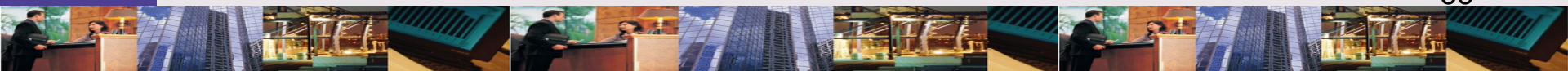
CUSTOMER PERSPECTIVE

- ✧ Source of revenue component within the financial perspective
 - ✧ Core objectives & measures
 - ✧ Customer value
 - ✧ Difference between what customers receive and what they have given up
 - ✧ Delivery reliability



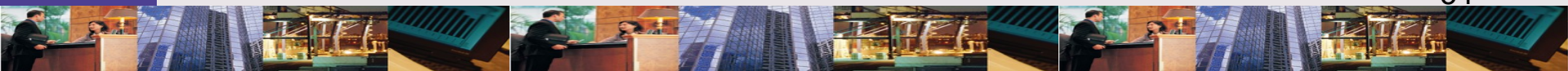
PROCESS PERSPECTIVE

- ✧ Process value chain made up of 3 processes
 - ✧ Innovation process
 - ✧ Operations process
 - ✧ Cycle time & velocity
 - ✧ Manufacturing cycle efficiency
 - ✧ Day-by-hour report
 - ✧ Post sales service process



LEARNING & GROWTH PERSPECTIVE

- ✧ Source of capabilities that enable the accomplishment of other 3 perspectives
 - ✧ Employee capabilities
 - ✧ Motivation, empowerment, alignment
 - ✧ Information systems capabilities



CHAPTER 16

THE END